

Equation of motion

(4-state: no disturbance state)

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\},$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\},$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired trajectory, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired step size.

Measurement

(d -step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1 - \lambda_c) \delta x[k]\}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k] f_T[k]$$

d -step measurement delay:

$$\delta x[k] = \delta x[k-d] + \sum_{i=1}^d \Delta x_d[k-i] - \sum_{i=1}^d a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\}$$

Ideal control effort (with d -step measurement delay):

$$\bar{\bar{f}}_{dx}[k] = a_x^{-1}[k]\left\{ \underbrace{\left[\Delta x_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta x_d[k-i] \right]}_{\Delta x_d^d[k]} + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] \right\}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\left\{ a_x[k] f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_x[k-i] f_T[k-i] \right\}}_{\delta x_T^d[k]}$$

Implementable control effort

$$f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k] [f_{dx}[k] - \bar{\bar{f}}_{dx}[k]] - \delta x_T^d[k]$$

$$a_x[k] \bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right]$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &= e_{a_x}[k] f_{dx}[k] + (1 - \lambda_c) n_x[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_x[k-i] - \hat{a}_x[k-i]) f_{dx}[k-i] \end{aligned}$$

Parameter model (para: gain state $a_x \rightarrow$ wall-shape constant θ):

$$a_x[k] = a(\bar{h}[k]; \theta) = \frac{a_{\text{nom}}}{c(\bar{h}[k]; \theta)}, \quad a_{\text{nom}} = \Delta t / \gamma_N$$

$$c(\bar{h}; \theta) = 1 + \theta \varphi(\bar{h}), \quad \varphi = \begin{cases} 1/(\bar{h} - 1), & z \text{ (pole)} \\ 1/\bar{h}, & x, y \end{cases}, \quad \varphi' = -\varphi^2, \quad \varphi'' = 2\varphi^3$$

$$\frac{\partial a}{\partial \theta} = -\frac{a_{\text{nom}} \varphi}{c^2} = -\frac{a}{c} \varphi$$

$$a' = \frac{da}{dh} = \frac{1}{R} \frac{da}{d\bar{h}} = \frac{a_{\text{nom}} \theta \varphi^2}{R c^2}, \quad a'' = \frac{1}{R^2} \frac{a_{\text{nom}}}{c^2} \left(\frac{2\theta^2 \varphi^4}{c} - 2\theta \varphi^3 \right)$$

Parameter / estimator / error (θ constant):

$$\begin{cases} \theta[k+1] = \theta[k] \\ \hat{\theta}[k+1] = \hat{\theta}[k] \end{cases} \quad e_\theta[k] = \theta - \hat{\theta}[k]$$

Gain error (linearized about $\hat{\theta}$):

$$e_{a_x}[k] = a(\bar{h}; \theta) - a(\bar{h}; \hat{\theta}) = \left. \frac{\partial a}{\partial \theta} \right|_{\bar{h}[k]} e_\theta[k] + O(e_\theta^2)$$

Past gain errors (θ constant $\Rightarrow$ no $i e_{\delta a_x}$, no $\sum \delta a_{ram}$; cf 6state Page 3):

$$a_x[k-i] - \hat{a}_x[k-i] \approx \left. \frac{\partial a}{\partial \theta} \right|_{\bar{h}[k-i]} e_\theta[k]$$

$$a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] \approx e_{a_x}[k] \underbrace{\left\{ f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i] \right\}}_{F_{dx}[k]} + (1 - \lambda_c) n_x[k]$$

$$\boxed{\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \underbrace{\left\{ \delta x_T^d[k] + (1 - \lambda_c) n_x[k] \right\}}_{\varepsilon[k]}}$$

(No $dF_{dx} e_{\delta a_x}$ term and no $\delta x_{\delta af}^d$ term: the para model has no δa_x state and no δa_{ram} .)

State equation

$$\delta x_1[k+1] = \delta x_2[k]$$

$$\delta x_2[k+1] = \delta x_3[k]$$

$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k], \quad e_{a_x} = \frac{\partial a}{\partial \theta} e_\theta$$

$$\theta[k+1] = \theta[k]$$

Output equation

(y_2 nonlinear in θ)

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} = a(\bar{h}[k-d]; \theta) + \underbrace{n_a[k]}_{r_2[k]} \end{cases}$$

(d -step delay absorbed by evaluating $a(\cdot)$ at $\bar{h}[k-d]$; hence $r_2 = n_a$ only, cf 6state $r_2 = n_a - \sum \delta a_{ram}$.)

$$\boxed{\mathbf{h}(\mathbf{x}) = \begin{bmatrix} \delta x_1 \\ a(\bar{h}[k-d]; \theta) \end{bmatrix}, \quad \mathbf{H}[k] = \frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \left. \frac{\partial a}{\partial \theta} \right|_{\bar{h}[k-d]} \end{bmatrix}}$$

$$\mathbf{y}[k] = \mathbf{h}(\mathbf{x}[k]) + \mathbf{r}[k]$$

Closed-loop estimator (EKF)

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta \hat{x}_1[k], \quad e_{\theta}[k] = \theta - \hat{\theta}[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$e_{y_2}[k] = y_2[k] - \hat{y}_2[k] = a_{xm}[k] - a(\bar{h}[k-d]; \hat{\theta}[k]) = \left. \frac{\partial a}{\partial \theta} \right|_{\bar{h}[k-d]} e_{\theta}[k] + \underbrace{n_a[k]}_{r_2[k]}$$

$$\delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11}[k] e_{y_1}[k] + \ell_{12}[k] e_{y_2}[k]$$

$$\delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21}[k] e_{y_1}[k] + \ell_{22}[k] e_{y_2}[k]$$

$$\delta \hat{x}_3[k+1] = \lambda_c \cdot \delta \hat{x}_3[k] + \ell_{31}[k] e_{y_1}[k] + \ell_{32}[k] e_{y_2}[k]$$

$$\hat{\theta}[k+1] = \hat{\theta}[k] + \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]$$

Error dynamics

$$(e_{a_x} = \frac{\partial a}{\partial \theta} e_{\theta})$$

$$e_{\delta x_1}[k+1] = e_{\delta x_2}[k] - \ell_{11} e_{\delta x_1}[k] - \ell_{12} \frac{\partial a}{\partial \theta} e_{\theta}[k]$$

$$e_{\delta x_2}[k+1] = e_{\delta x_3}[k] - \ell_{21} e_{\delta x_1}[k] - \ell_{22} \frac{\partial a}{\partial \theta} e_{\theta}[k]$$

$$e_{\delta x_3}[k+1] = \lambda_c e_{\delta x_3}[k] - F_{dx} \frac{\partial a}{\partial \theta} e_{\theta}[k] - \ell_{31} e_{\delta x_1}[k] - \ell_{32} \frac{\partial a}{\partial \theta} e_{\theta}[k] - \underbrace{\varepsilon[k]}_{q_3[k]}$$

$$e_{\theta}[k+1] = e_{\theta}[k] - \ell_{41} e_{\delta x_1}[k] - \ell_{42} \frac{\partial a}{\partial \theta} e_{\theta}[k]$$

(No q on e_{θ} : θ is a constant $\Rightarrow Q_{\theta} = 0$.)

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \cdot \frac{\partial a}{\partial \theta} \big|_{\bar{h}[k]} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(cf 4state_del_hd: $\mathbf{F}_e(3,4) = -F_{dx}$, $\mathbf{F}_e(4,4) \in \{1, \lambda_c\}$; here the gain coupling carries the Jacobian $\partial a / \partial \theta$, and $\mathbf{F}_e(4,4) = 1$.)

$$\mathbf{e}[k+1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H}[k] \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L} \mathbf{r}[k]$$

Process / measurement covariance

$$Q_{33} = 4k_B T \hat{a}_x (1 + (1 - \lambda_c)^2 d) + (1 - \lambda_c)^2 \sigma_n^2, \quad \mathbf{Q} = \text{diag}(0, 0, Q_{33}, Q_\theta)$$

$$Q_\theta = 0 \quad (\theta \text{ constant}) \iff \text{4state } Q_{44} = \text{Var}(\delta a_{ram}) = \left(\frac{\hat{a}_x K_h}{R}\right)^2 C_{dx} \sigma_{\delta h}^2 \quad (\text{eliminated})$$

$$R_{11} = \sigma_n^2, \quad R_{22} = K_{\text{var}} \text{IF}_{\text{eff}}(\hat{a}_x + \xi)^2 + d \cdot \text{Var}(\delta a_{inc})$$

Adaptive measurement gate

(y_2 dropped, $\mathbf{H} \rightarrow \mathbf{H}(1, :)$, $\mathbf{R} \rightarrow \sigma_n^2$ when any holds)

$$G_1 : t < t_{\text{warm}}; \quad G_2 : \hat{\sigma}_{\delta x_r}^2[k] \leq C_n \sigma_n^2$$

(no near-wall $\bar{h}$ gate: near-wall a_{xm} used directly.)

Caveats (numeric)

$$(i) \text{ form misfit (noise-free): } \text{RMS}[a'_{\text{model}}/a'_{\text{true}} - 1] \approx \begin{cases} 1\%, & z \text{ (pole)} \\ 14\text{--}22\%, & x, y \end{cases}$$

$$(ii) a_{xm} = (1 + \beta) a_x \Rightarrow \hat{\theta} \text{ absorbs } \beta \text{ as scale; } \beta|_{\bar{h} \rightarrow 1} \approx +0.11, \quad \hat{a}'/a' - 1 \approx -\beta'/K_h \text{ (suppressed)}$$